Technology Stack

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	***Frontend / Client-Side*** Frontend/Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, Jinja2 Templating	Lightweight web interface builder; clean response display and user-friendly interaction format.

2	***Backend / Server-Side*** Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3.10+, FastAPI, Uvicorn Server	High-performance asynchronous REST API handling, fast route processing, and lightweight server execution.
3	***Database / Data Storage*** Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Stateless / In-Memory Session Storage	No explicit external database connection; suitable for lightweight architecture setup and processing.
4	***Cloud / Hosting / Deployment*** Cloud/Hosting /Deployment (e.g., AWS, Azure, Heroku, Docker)	Local Execution Environment (Mac M1/CPU compatibility)	Low-footprint application infrastructure running locally on standard computing hardware.

5	***Version Control & CI/CD*** Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git/GitHub	Code tracking, repository management, and structured project folder collaboration support.

6	***Third-Party APIs / Other Tools*** Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google Gemini 1.5 Pro API & LaMini-Flan-T5-783M Model	Q&A, Summarization, Quiz Generation, Learning Path Recommendations, and lightweight local concept explanations.